

Differentiation 2 HW

Consolidation

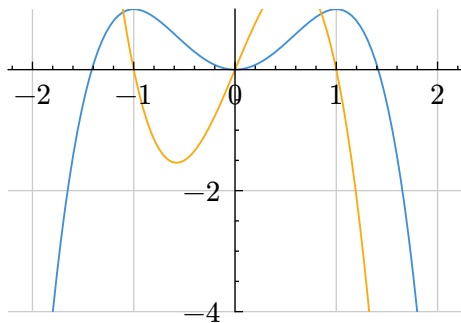
4. $A = p'(x)$

$$B = g'(x)$$

$$C = q'(x)$$

$$D = f'(x)$$

5.



Exam Questions

1. $f(x) = 6x^3 - 5x$

$$\Rightarrow f'(x) = 18x^2 - 5$$

$$\Rightarrow f''(x) = 36x$$

$$f''(2) = 36 \times 2 = 72$$

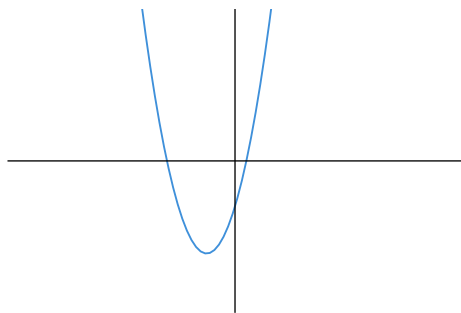
2. $f(x) = \frac{2}{3}x^3 + \frac{5}{2}x^2 - 3x + 7$

$$\therefore f'(x) = 2x^2 + 5x - 3 = (2x - 1)(x + 3)$$

When the gradient of the curve $f(x)$ is positive, $f'(x) > 0$.

$$\text{So } (2x - 1)(x + 3) > 0.$$

A graph of $f'(x)$



Therefore $x < -3$ or $x > \frac{1}{2}$

$$x = \{x : x < -3\} \cup \{x : x > \frac{1}{2}\}$$

3. A curve has equation $y = f(x)$, where $f(x)$ is a quadratic polynomial in x .
The curve passes through $(0, 3)$ and $(4, -13)$.
At the point where $x = 3$, the gradient of the curve is -2 .

Therefore $f(0) = 3$, $f(4) = -13$, and $f'(3) = -2$.

Let $f(x) = ax^2 + bx + c \Rightarrow f'(x) = 2ax + b$.

$$f(0) = 3$$

$$\therefore c = 3.$$

$$f(4) = a(4)^2 + b(4) + 3 = 16a + 4b + 3 = -13$$

$$\therefore 4a + b = -4$$

$$\text{And } f'(3) = 6a + b = -2$$

Then we can form two simultaneous equations from $f'(3)$ and $f(4)$:

$$\begin{cases} 6a + b = -2 \\ 4a + b = -4 \end{cases}$$

$$\therefore 2a = 2$$

$$a = 1$$

$$\therefore b + 6 = -2$$

$$b = -8$$

Therefore, $f(x) = x^2 - 8x + 3$
